

Determine and simplify the 8th term in the binomial expansion of $\left(x - \frac{2}{x^3}\right)^{10}$.

Solution

$$t_8 = {}_{10}C_7 (x)^3 \left(-\frac{2}{x^3}\right)^7 \quad 2 \text{ marks (1 mark for } {}_{10}C_7; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$= 120 \left(x^3\right) \left(-\frac{128}{x^{21}}\right)$$

$$= -\frac{15\,360}{x^{18}}$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient; $\frac{1}{2}$ mark for exponent)

3 marks

Question 2

R10

The temperature of hot chocolate is recorded as it cools down. The data shows that the temperature cools according to the equation:

$$T = 82(0.87)^t + 16$$

where T is the temperature of the hot chocolate, in degrees Celsius after t minutes and

t is the time, in minutes, after the hot chocolate is made.

In order to avoid burns, hot chocolate should not be served at a temperature higher than 71°C . Determine, algebraically, the amount of time it will take the hot chocolate to reach this temperature.

Solution

$$71 = 82(0.87)^t + 16$$

$$55 = 82(0.87)^t$$

$$\frac{55}{82} = 0.87^t$$

$$\log\left(\frac{55}{82}\right) = t \log(0.87)$$

$\frac{1}{2}$ mark for applying logarithms
1 mark for power law

$$t = \frac{\log\left(\frac{55}{82}\right)}{\log(0.87)}$$

$$t = 2.867\ 874$$

$$t = 2.868 \text{ minutes}$$

$\frac{1}{2}$ mark for evaluating quotient of logarithms

2 marks

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$3\sin^2\theta + 6\sin\theta + 2 = 0$$

Solution

$$\sin\theta = \frac{-6 \pm \sqrt{6^2 - 4(3)(2)}}{2(3)}$$

½ mark for substitution into quadratic formula

$$\sin\theta = \frac{-6 \pm \sqrt{12}}{6}$$

½ mark for solving for $\sin\theta$

$$\sin\theta = -0.422\ 649\dots$$

$$\sin\theta = -1.577\ 350\dots$$

$$\theta_r = 0.436\ 367\dots$$

no solution

$$\theta = 3.578, 5.847$$

2 marks for solving for θ

(½ mark for each value; 1 mark for indicating no solution)

3 marks

Question 4

P3

A committee of 4 people is to be selected from 6 high school students and 5 middle years students.

Determine the number of possible committees if the committee must have at least 3 high school students.

Solution

Case 1: ${}_6C_3 \cdot {}_5C_1 = 100$

1 mark for Case 1

Case 2: ${}_6C_4 \cdot {}_5C_0 = 15$

1 mark for Case 2

$100 + 15 = 115$ possible committees

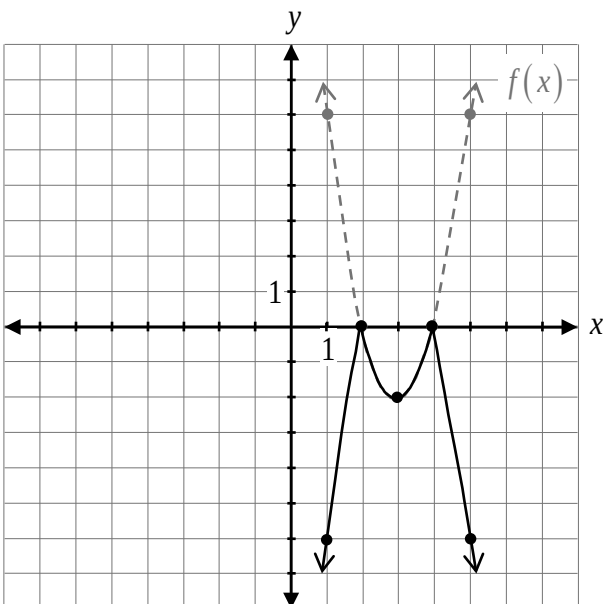
1 mark for addition of cases

3 marks

Note:

- ${}_5C_0$ does not need to be shown.

Given the graph of $y = f(x)$, sketch the graph of $y = -|f(x)|$.

Solution

1 mark for absolute value
1 mark for vertical reflection

2 marks

Describe the transformations used to obtain the graph of the function $y = 4f\left(\frac{x}{2}\right) - 3$ from the graph of $y = f(x)$.

Solution

The graph of $y = f(x)$ is vertically stretched by a factor of 4, horizontally stretched by a factor of 2, and translated down 3 units.

1 mark for vertical stretch

1 mark for horizontal stretch

1 mark for vertical translation

3 marks

Note:

- Deduct a maximum of 1 mark for concept error of incorrect order of vertical transformations.

If $\log 4 = m$ and $\log 3 = n$, express $\log 48$ in terms of m and n .

Solution

$$\log 48 = \log(4^2 \cdot 3)$$

$$= 2\log 4 + \log 3$$

$$= 2m + n$$

1 mark for power law

1 mark for product law

2 marks

State the domain and range of the radical function, $f(x) = -3\sqrt{x-8} + 1$.

Solution

Domain: $[8, \infty)$

1 mark for domain

Range: $(-\infty, 1]$

1 mark for range

2 marks

Explain why the graph of $y = \tan x$ does not have an amplitude.

Solution

The range of $y = \tan x$ is all real numbers.

1 mark**or**

As the graph of $y = \tan x$ approaches the vertical asymptotes, the function value approaches positive and negative infinity.

Prove the following identity for all permissible values of θ .

$$\frac{\sec \theta - \sin^2 \theta \sec \theta}{\tan \theta \sin \theta} = \csc^2 \theta - 1$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\frac{1}{\cos \theta} - \frac{\sin^2 \theta}{\cos \theta}}{\left(\frac{\sin \theta}{\cos \theta}\right)(\sin \theta)}$ $\frac{1 - \sin^2 \theta}{\cancel{\cos \theta}}$ $\frac{\sin^2 \theta}{\cancel{\cos \theta}}$ $\frac{\cos^2 \theta}{\sin^2 \theta}$ $\cot^2 \theta$ $\csc^2 \theta - 1$	$\csc^2 \theta - 1$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\frac{\sec \theta (1 - \sin^2 \theta)}{\left(\frac{\sin \theta}{\cos \theta}\right)(\sin \theta)}$ $\frac{\left(\frac{1}{\cos \theta}\right)(\cos^2 \theta)}{\frac{\sin^2 \theta}{\cos \theta}}$ $\frac{\cos^2 \theta}{\cancel{\cos \theta} \frac{\sin^2 \theta}{\cancel{\cos \theta}}}$ $\cot^2 \theta$ $\csc^2 \theta - 1$	$\csc^2 \theta - 1$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 11

R11

Given that $x = 3$ is one of the zeros of $p(x) = x^3 - 7x - 6$, express $p(x)$ in completely factored form.

Solution

$$\begin{array}{r|rrrr} 3 & 1 & 0 & -7 & -6 \\ & & 3 & 9 & 6 \\ \hline & 1 & 3 & 2 & 0 \end{array}$$

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x - 3)(x^2 + 3x + 2)$$

$\frac{1}{2}$ mark for factor of $(x - 3)$

$$p(x) = \underline{(x - 3)(x + 2)(x + 1)}$$

$\frac{1}{2}$ mark for other consistent factors

2 marks

Solve, algebraically.

$${}_{n+1}P_2 = 6$$

Solution

$$\frac{(n+1)!}{(n+1-2)!} = 6$$

½ mark for substitution

$$\frac{(n+1)!}{(n-1)!} = 6$$

$$\frac{(n+1)(n)\cancel{(n-1)!}}{\cancel{(n-1)!}} = 6$$

1 mark for factorial expansion

½ mark for simplification of factorials

$$n^2 + n = 6$$

$$n^2 + n - 6 = 0$$

$$(n+3)(n-2) = 0$$

$$\cancel{n+3} \quad n = 2$$

½ mark for the permissible value of n

½ mark for showing the rejection of the extraneous root

3 marks

Express $\log(3x - 1) - \log x + \log 9$ as a single logarithm.

Solution

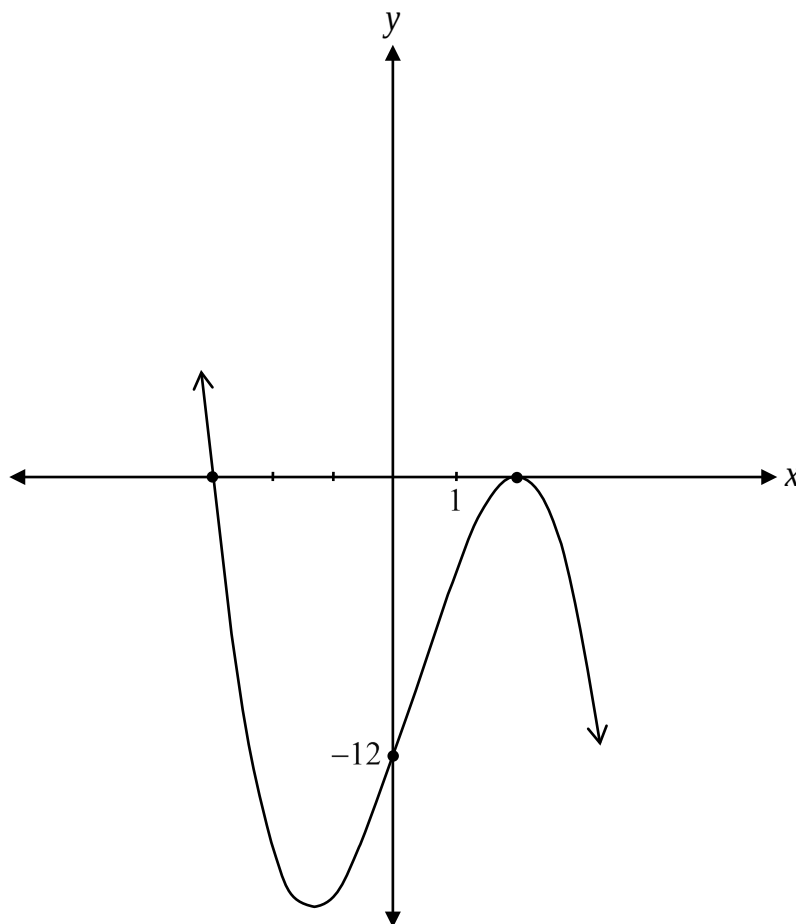
$$\log\left(\frac{9(3x - 1)}{x}\right)$$

1 mark for product law
1 mark for quotient law

2 marks

Sketch the graph of $P(x) = -(x-2)^2(x+3)$.

Solution



1 mark for x-intercepts

½ mark for y-intercept

1 mark for multiplicity of 2 at $x = 2$

½ mark for end behaviour

3 marks

Question 15**R1**

Given $f(x) = -2x - 5$ and $g(x) = x + 7$, identify the equation of $h(x) = f(g(x))$.

a) $h(x) = -2x^2 - 35$

b) $h(x) = -2x - 19$

c) $h(x) = -2x^2 - 19x - 35$

d) $h(x) = -2x + 2$

Question 16**R6**

The function $y = f(x)$ has a domain of $[2, 5]$ and a range of $[-6, 3]$. Identify the range of the function $y = f^{-1}(x)$.

a) $[-6, 3]$

b) $[-5, 2]$

c) $[-3, 6]$

d) $[2, 5]$

Question 17**R14**

Identify the function that has a graph with a point of discontinuity (hole) at $x = -3$.

a) $y = \frac{x+3}{x^2-9}$

b) $y = \frac{x-3}{x^2-9}$

c) $y = \frac{x^2-9}{x-3}$

d) $y = \frac{x^2+9}{x+3}$

Question 18

R7

Identify the value of $\ln e$.

a) 0

b) $\log e$

c) 1

d) e

Question 19

R11

When the polynomial function, $p(x)$, is divided by $(x-4)$ the remainder is 17. Identify which statement is true.

a) $p(4) = 17$

b) $p(-4) = 17$

c) $p(4) = 0$

d) $p(-4) = 0$

Question 20

P2

Identify the expression equivalent to $\frac{(n-6)!}{(n-4)!}$.

a) $(n-4)(n-5)$

b) $\frac{1}{(n-4)(n-5)}$

c) $(n-6)(n-5)$

d) $\frac{1}{(n-6)(n-5)}$

Question 21

T3

Identify the quadrant in which θ terminates if $\sec \theta = -\frac{4}{3}$ and $\sin \theta > 0$.

a) I

b) II

c) III

d) IV

Question 22

T1

Identify the expression that represents all angles that are coterminal with $\frac{\pi}{3}$.

a) $\frac{\pi}{3} + \pi k, k \in \mathbb{Z}$

b) $\frac{\pi}{3} + \pi k, k \in \mathbb{R}$

c) $\frac{\pi}{3} + 2\pi k, k \in \mathbb{Z}$

d) $\frac{\pi}{3} + 2\pi k, k \in \mathbb{R}$

Given $f(x) = \frac{1}{2}x - 3$, state the coordinates of an invariant (unchanged) point when sketching the graph of $y = \sqrt{f(x)}$.

Solution

$$\frac{1}{2}x - 3 = 0$$

$$\frac{1}{2}x = 3$$

$$x = 6$$

$$(6, 0)$$

1 mark**or**

$$\frac{1}{2}x - 3 = 1$$

$$\frac{1}{2}x = 4$$

$$x = 8$$

$$(8, 1)$$

Solve, algebraically.

$$125^{3x+4} = \left(\frac{1}{5}\right)^x$$

Solution

$$\left(5^3\right)^{3x+4} = \left(5^{-1}\right)^x$$

1 mark for changing to a common base (½ mark for each side)

$$5^{9x+12} = 5^{-x}$$

½ mark for exponent law

$$9x + 12 = -x$$

½ mark for equating exponents

$$10x = -12$$

$$x = -\frac{12}{10}$$

2 marks

or

$$x = -\frac{6}{5}$$

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ and $\cos \theta = -\frac{\sqrt{2}}{2}$, justify that the value of $\tan(2\theta)$ is undefined.

Solution**Method 1**

$$\begin{aligned}\tan(2\theta) &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \\ &= \frac{2(1)}{1-1} \\ &= \frac{2}{0}\end{aligned}$$

1 mark for value of $\tan \theta$

$\therefore$ undefined

1 mark for justification

2 marks

Method 2

$$\tan \theta = 1$$

$$\theta = \frac{5\pi}{4}$$

1 mark for value of θ

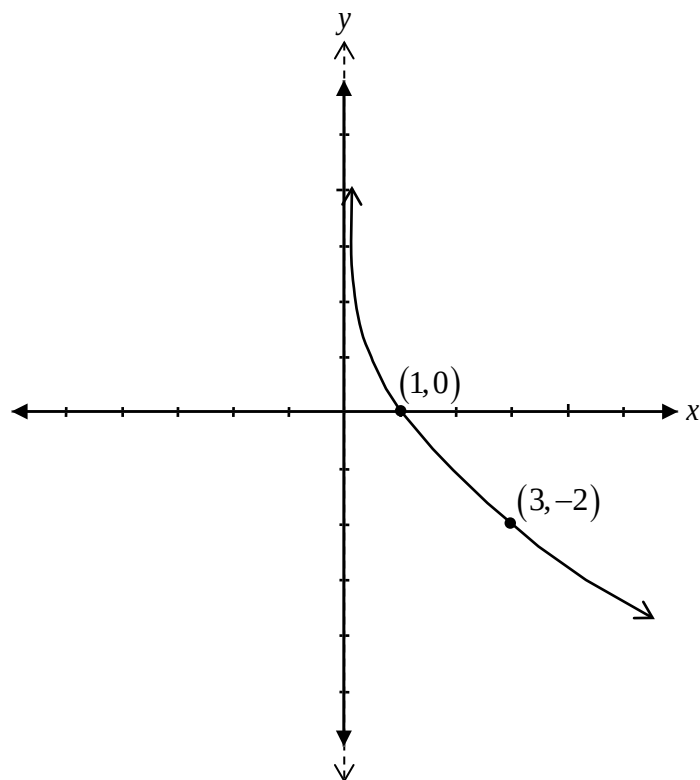
$$\begin{aligned}\tan(2\theta) &= \tan\left(2\left(\frac{5\pi}{4}\right)\right) \\ &= \tan \frac{5\pi}{2}\end{aligned}$$

$\therefore$ undefined

1 mark for justification

2 marks

Sketch the graph of $y = -2\log_3 x$.

Solution

1 mark for asymptotic behaviour
approaching $x = 0$

1 mark for vertical reflection

1 mark for vertical stretch

3 marks

Evaluate.

$$\cos\left(\pi \cdot \sin\left(-\frac{\pi}{6}\right)\right)$$

Solution

$$\cos\left(\pi\left(-\frac{1}{2}\right)\right)$$

1 mark for $\sin\left(-\frac{\pi}{6}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

$$\cos\left(-\frac{\pi}{2}\right)$$

1 mark for consistent value

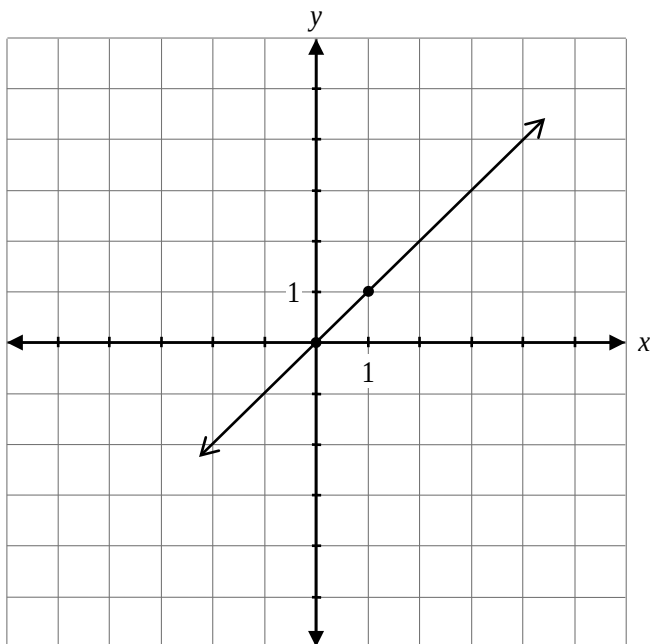
0

2 marks

- a) Given $f(x) = -x + 2$, sketch the graph of $h(x) = f(f(x))$.
- b) Explain why the domain of $h(x) = f(f(x))$ does not have any restrictions.

Solution

a)

**1 mark**

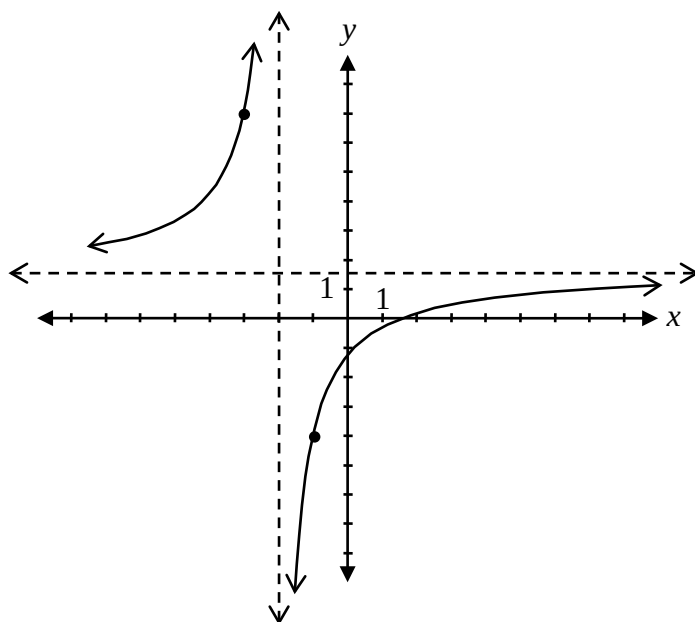
- b) Since the domain of $y = f(x)$ does not have any restrictions, then the domain of the composition does not have any restrictions.

1 mark

a) Sketch the graph of $f(x) = \frac{3x-5}{2x+4}$.

b) State the range of $f(x)$.

Solution



1 mark for asymptotic behaviour approaching $x = -2$

1 mark for asymptotic behaviour approaching $y = \frac{3}{2}$

$\frac{1}{2}$ mark for graph right of $x = -2$

$\frac{1}{2}$ mark for graph left of $x = -2$

3 marks

b) Range: $\left\{ y \in \mathbb{R} \mid y \neq \frac{3}{2} \right\}$

1 mark for range consistent with graph

1 mark

Determine, algebraically, the equation of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 4
- $p(x)$ has a zero at 3 with a multiplicity of 2
- $p(x)$ has zeroes at -1 and -2
- $p(x)$ passes through the point $(2, 24)$.

Solution

$$p(x) = a(x+1)(x+2)(x-3)^2$$

$\frac{1}{2}$ mark for factors of $p(x)$

$\frac{1}{2}$ mark for multiplicity of 2 at $x = 3$

$$24 = a(2+1)(2+2)(2-3)^2$$

$\frac{1}{2}$ mark for substitution of $p(2) = 24$

$$24 = 12a$$

$$a = 2$$

$\frac{1}{2}$ mark for consistent value of a

$$p(x) = \underline{2(x+1)(x+2)(x-3)^2}$$

2 marks

Verify, by substitution, that the equation $\frac{\cos \theta + \cot \theta}{\cot \theta} = 1 + \sin \theta$ is true for $\theta = \frac{2\pi}{3}$.

Solution

Left-Hand Side	Right-Hand Side
$\frac{\cos\left(\frac{2\pi}{3}\right) + \cot\left(\frac{2\pi}{3}\right)}{\cot\left(\frac{2\pi}{3}\right)}$	$1 + \sin\left(\frac{2\pi}{3}\right)$ ½ mark for substitution
$\frac{-\frac{1}{2} - \frac{1}{\sqrt{3}}}{-\frac{1}{\sqrt{3}}}$	$1 + \frac{\sqrt{3}}{2}$ 1½ marks for exact values (½ mark for each)
$\frac{-\frac{\sqrt{3}-2}{2\sqrt{3}}}{-\frac{1}{\sqrt{3}}}$	$\frac{2}{2} + \frac{\sqrt{3}}{2}$
$\frac{-\sqrt{3}-2}{-2}$	$\frac{2+\sqrt{3}}{2}$
$\frac{-(\sqrt{3}+2)}{-2}$	$\frac{\sqrt{3}+2}{2}$ 1 mark for simplification (½ mark for LHS; ½ mark for RHS)
$\frac{\sqrt{3}+2}{2}$	3 marks

Note:

- Deduct a maximum of 1 mark if student proves identity without substitution of $\theta = \frac{2\pi}{3}$.

Jyugo was asked to state the equation of the vertical asymptote(s) on the graph of $f(x) = \frac{3x-15}{x^2-5x}$.

His solution:

$$f(x) = \frac{3(x-5)}{x(x-5)}$$

$$x=0 \quad x=5$$

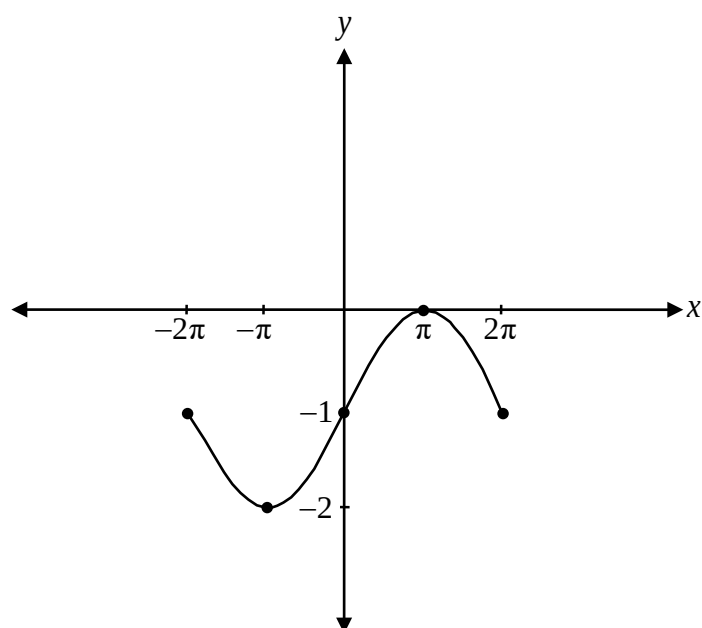
Explain why his solution is incorrect.

Solution

His solution at $x=5$ is the x -value of a point of discontinuity (hole), not a vertical asymptote.

1 mark

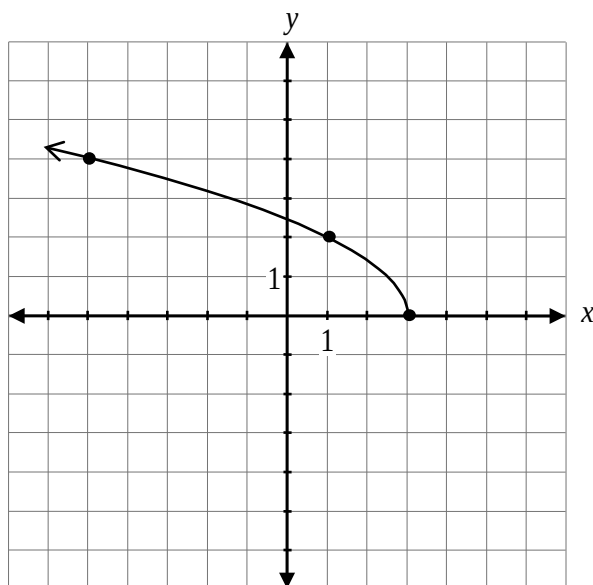
Sketch the graph of $y = \sin\left(\frac{1}{2}x\right) - 1$ over the interval $[-2\pi, 2\pi]$.

Solution

1 mark for shape of $y = \sin x$
1 mark for period
1 mark for vertical translation

3 marks

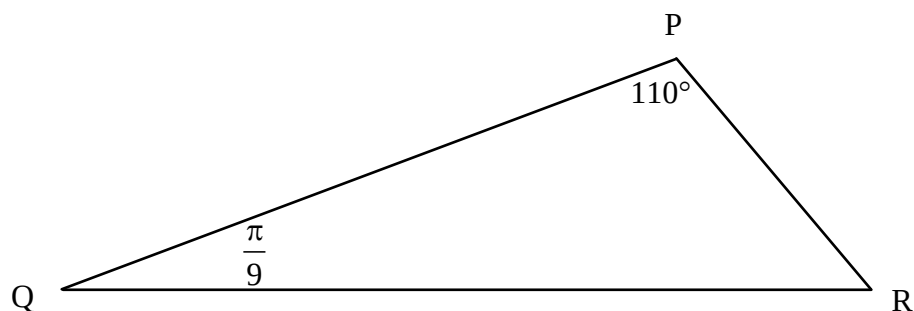
Sketch the graph of $y = \sqrt{-2x + 6}$.

Solution

- 1 mark for shape of a radical function
- 1 mark for horizontal reflection
- 1 mark for horizontal compression
- 1 mark for horizontal translation

4 marks

Determine the measure of angle R, in radians.

**Solution**

$$(110)\left(\frac{\pi}{180}\right) = \frac{11\pi}{18} \quad \text{1 mark for conversion}$$

$$\angle R = \pi - \left(\frac{\pi}{9} + \frac{11\pi}{18}\right)$$

$$\angle R = \frac{18\pi}{18} - \left(\frac{2\pi}{18} + \frac{11\pi}{18}\right)$$

$$\angle R = \frac{5\pi}{18} \quad \text{1 mark for angle R}$$

2 marks

Given $f(x) = \sqrt{x-4}$, state the equation of the resulting function, $g(x)$, after a reflection over the x -axis.

Solution

$$g(x) = -\sqrt{x-4}$$

1 mark for vertical reflection

1 mark

a) Determine the coterminal angle of $\frac{29\pi}{12}$ over the interval $[0, 2\pi]$.

b) Determine the exact value of $\sin\left(\frac{29\pi}{12}\right)$.

Solution

a) $\frac{5\pi}{12}$

1 mark

b)

$$\sin\left(\frac{5\pi}{12}\right) = \sin\left(\frac{\pi}{6} + \frac{\pi}{4}\right)$$

1 mark for substitution into correct identity

$$= \sin\frac{\pi}{6}\cos\frac{\pi}{4} + \cos\frac{\pi}{6}\sin\frac{\pi}{4}$$

$$= \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

2 marks ($\frac{1}{2}$ mark for each exact value)

$$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4}$$

$$= \frac{\sqrt{2} + \sqrt{6}}{4}$$

3 marks

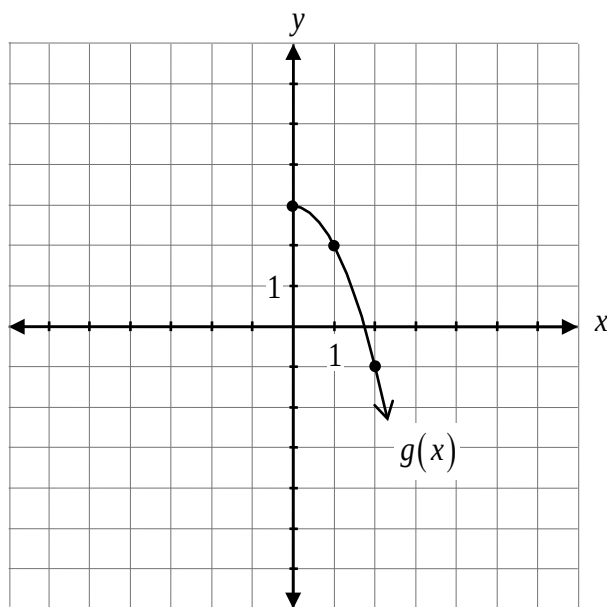
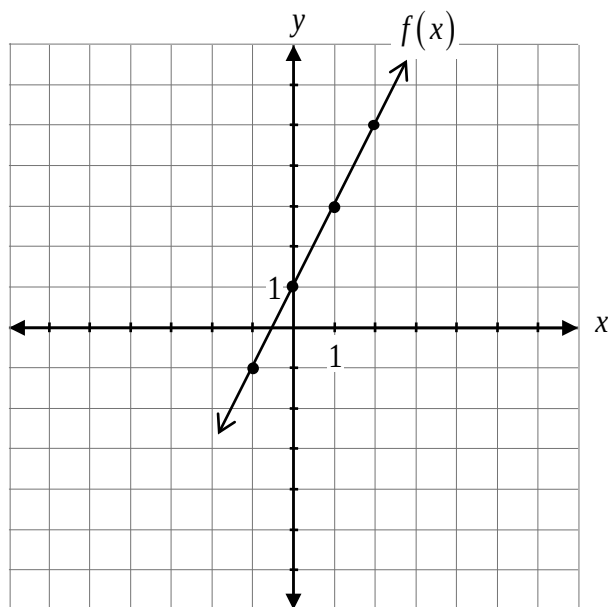
or

$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

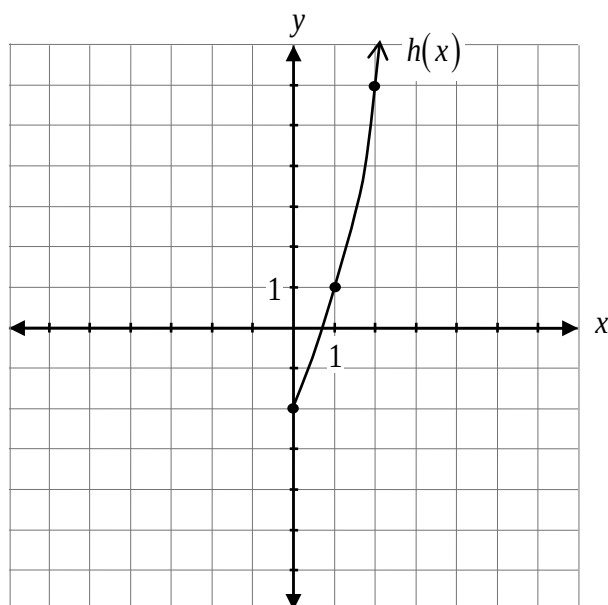
Note:

- Other combinations are possible.

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) - g(x)$.



Solution



1 mark for operation of subtraction
1 mark for restricted domain

2 marks

Determine an equation of a sinusoidal function that has the following characteristics:

- an amplitude of 3
- a period of 6
- a minimum value of -5

Solution

$$y = A \sin(Bx) + D$$

$$y = 3 \sin\left(\frac{\pi}{3}x\right) - 2$$

or

$$y = A \cos(Bx) + D$$

$$y = 3 \cos\left(\frac{\pi}{3}x\right) - 2$$

1 mark for value of A

1 mark for value of B

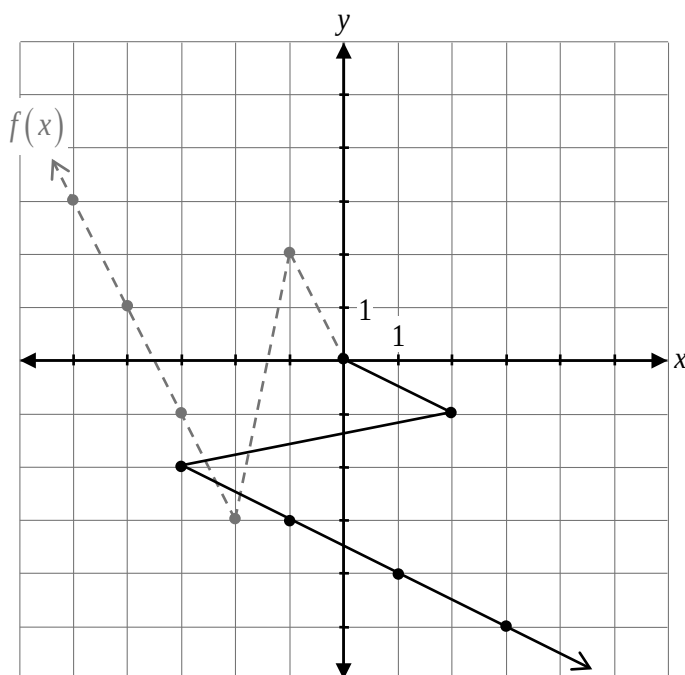
1 mark for value of D

3 marks

Note:

- Other equations are possible.

Given the graph of $y = f(x)$, sketch the graph after a reflection over the line $y = x$.

Solution

1 mark for reflection over the line $y = x$

1 mark

a) Determine the remainder when $3x^3 + 5x^2 - 13x - 3$ is divided by $(x + 3)$.

b) Is $(x + 3)$ a factor of $3x^3 + 5x^2 - 13x - 3$? Explain your reasoning.

Solution

a) $3(-3)^3 + 5(-3)^2 - 13(-3) - 3$

1 mark for remainder theorem

$$3(-27) + 5(9) - 13(-3) - 3$$

$$0$$

or

-3	3	5	-13	-3
		-9	12	3
	3	-4	-1	0

1 mark for synthetic division
(or any equivalent strategy)

The remainder is 0.

1 mark

b) Yes, because the remainder is 0.

1 mark for explanation

1 mark

State the range, the y-intercept, and the equation of the asymptote of the exponential function,
 $f(x) = 3^{x-1} + 2$.

SolutionRange: $(2, \infty)$

1 mark for range

y-intercept: $\frac{7}{3}$

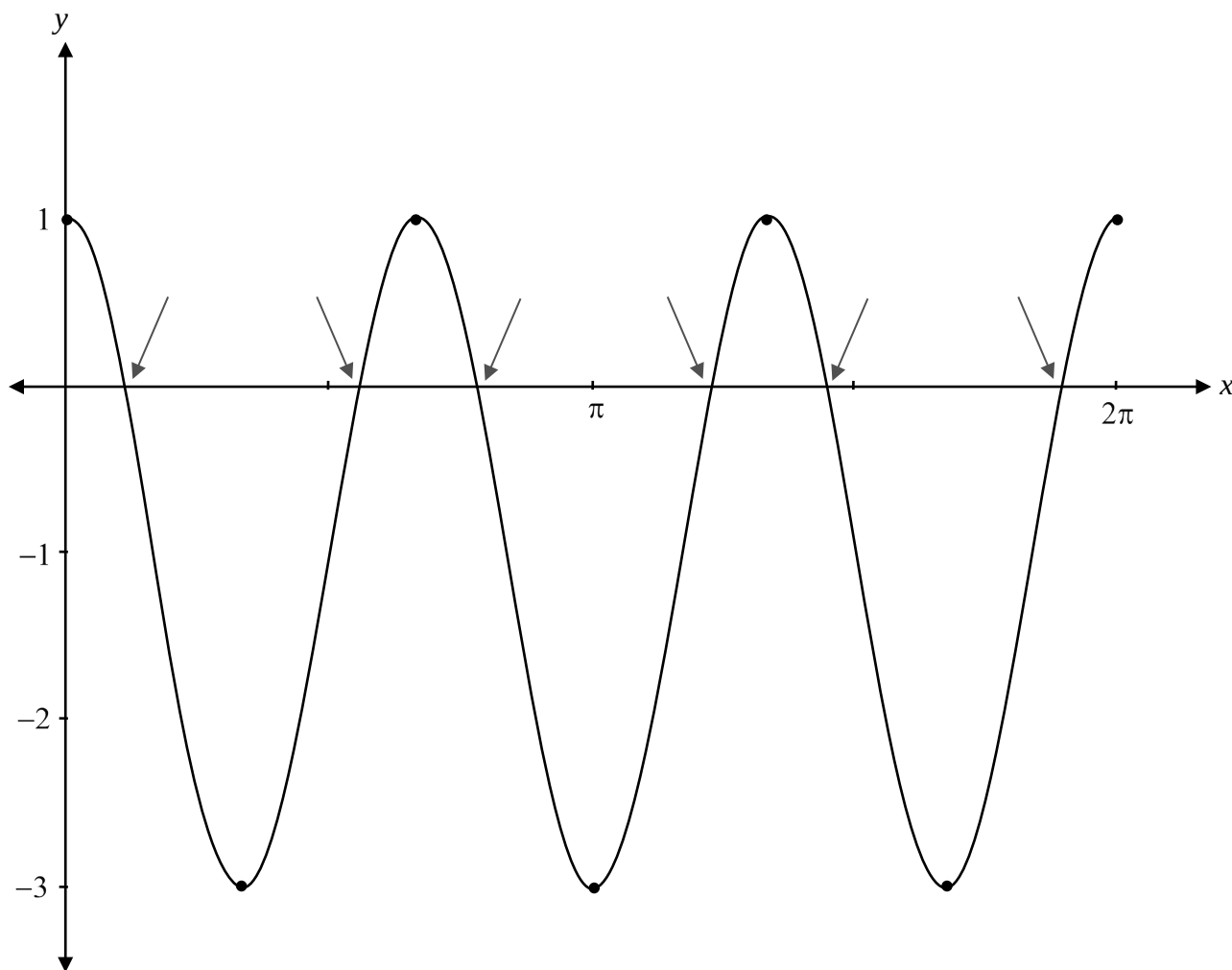
1 mark for y-intercept

Equation of the asymptote: $y = 2$

1 mark for equation of the asymptote

3 marks

The graph of $y = 2 \cos(3x) - 1$ below can be used to solve the equation $0 = 2 \cos(3x) - 1$ over the interval $[0, 2\pi]$. Indicate on the graph where to find at least one solution to the equation $0 = 2 \cos(3x) - 1$.

**Solution**

See arrows on graph above. 1 mark for indicating a solution on graph

1 mark

Note:

- Students do not need to show all six arrows.